Project Description: Homological Algebra Beyond Projective Space

This proposal involves research in algebraic geometry and commutative algebra with several connections to computation and combinatorics, as well as a wide array of broader impacts.

- § 1 **Results of Prior NSF Support.**
- § **?? Tropicalizations of Shimura Varieties**
- § **?? A General Theory of Tropical Shimura Varieties**
- § **?? Tropical Modular Interpretations**
- § **?? Computing Weight Zero Compactly Supported Cohomology of Shimura Varieties**
- § **?? Algebraic Structures on the Weight Zero Cohomology of Shimura Varieties**
- § 4 **Broader Impacts.**

1. Results of Prior NSF Support

During 2020-2022 the PI was supported by an NSF Postdoctoral Research Fellowship (NSF Grant No. MSPRF DMS-2002239, $150,000) titled *Asymptotic Syzygies in Algebraic Geometry.* In this period the PI made major contributions to extending our understanding of homological algebra on toric varieties (see Section 1.1) and the cohomology of moduli spaces (see Section 1.2). These works resulted in 5 new papers being posted to the arXiv [BCE+22, BHS24, BHS22, BBC23, BBC+24].

During 2015-2020 the PI was supported by an NSF Graduate Research Fellowship (NSF Grant No. DGE-1256259, $150,000) title *Syzygies in Algebraic Geometry.* In this period the PI made substantial contributions to commutative algebra and algebraic geometry, including furthering our understanding of syzygies on higher dimensional varieties and extending classical results in algebraic geometry and commutative algebra to finite fields. This resulted in the PI posting 8 new papers to the arXiv [ABLS20, BBB+17, BE19, BL20, BEGY20, BEGY21, Bru24, Bru22a].

Additionally, of the 12 new papers the PI posted to the arXiv during these periods of support 9 of these above-mentioned papers were accepted for publication, including in such journals as *Algebra & Number Theory* [BE19], *Geometry & Topology* [BBC+24], *Journal of Algebra* [BCE+22], and *Experimental Mathematics* [BEGY20]. The PI contributed to 4 open-source software packages, one public-facing database, and two articles for the *Notices of the AMS* [BBB21, Bru22b]. All of these are available via the software *Macaulay2* or the PI's website and Github. A more detailed summary of these results and their intellectual merits continues in the following sections.

1.1 Homological Algebra on Toric Varieties Introduced by Mumford, the Castelnuovo–Mumford Regularity of a projective variety $X \subset \mathbb{P}^r$ is a measure of the complexity of X given in terms of the vanishing of certain cohomology groups of X. Roughly speaking one should think about Castelnuovo–Mumford regularity as being a numerical measure of geometric complexity. Mumford was interested in such a measure as it plays a key role in constructing Hilbert and Quot schemes. In particular, being d-regular implies that $\mathcal{F}(d)$ is globally generated. However, Eisenbud and Goto showed that regularity is also closely connected to interesting homological properties.

Theorem 1.1. [EG84] *Let $\mathcal{F}$ be a coherent sheaf on $\mathbb{P}^r$ and $M = \bigoplus_{e\in\mathbb{Z}} H^0(\mathbb{P}^r, \mathcal{F}(e))$ the corresponding section ring. The following are equivalent: (1) M is d-regular; (2) $\beta_{p,q}(M) = 0$ for all $p \geq 0$ and $q > d + i$; (3) $M_{\geq d}$ has a linear resolution.*

Maclagan and Smith introduced multigraded Castelnuovo–Mumford regularity, where $\mathbb{P}^r$ can be replaced by any toric variety. Similarly to the definition in the classical setting multigraded Castelnuovo–Mumford regularity is defined in terms of the vanishing of cohomology groups, however, the multigraded Castelnuovo–Mumford regularity of a subvariety or module is not a single number, but an infinite subset of $\mathbb{Z}^r$. As an example, let us consider the case of products of projective spaces. Fixing a dimension vector $\mathbf{n} = (n_1, n_2, \ldots, n_r) \in \mathbb{N}^r$ we let $\mathbb{P}^{\mathbf{n}} := \mathbb{P}^{n_1} \times \mathbb{P}^{n_2} \times \cdots \times \mathbb{P}^{n_r}$ and

$S = \mathbb{K}[x_{i,j} \mid 1 \leq i \leq r, 0 \leq j \leq n_i]$ be the Cox ring of $\mathbb{P}^{\mathbf{n}}$ with the $\mathrm{Pic}(X) \cong \mathbb{Z}^r$-grading given by $\deg x_{i,j} = \mathbf{e}_i \in \mathbb{Z}^r$, where $\mathbf{e}_i$ is the i-th standard basis vector in $\mathbb{Z}^r$. Fixing some notation given $\mathbf{d} \in \mathbb{Z}^r$ and $i \in \mathbb{Z}_{\geq 0}$ we let $L_i(\mathbf{d}) \coloneqq \bigcup_{\mathbf{v} \in \mathbb{N}, |\mathbf{v}|=i} (\mathbf{d} - \mathbf{v}) + \mathbb{N}^r$. Note when $r = 2$ the region $L_i(\mathbf{d})$ looks like a staircase with $(i+1)$-corners. Roughly speaking we define regularity by requiring the i-th cohomology of certain twists of $\mathcal{F}$ to vanish on L_i.

Definition 1.2. [MS04, Definition 6.1] *A coherent sheaf $\mathcal{F}$ on $\mathbb{P}^{\mathbf{n}}$ is $\mathbf{d}$-regular if and only if $H^i(\mathbb{P}^{\mathbf{n}}, \mathcal{F}(\mathbf{e})) = 0$ for all $\mathbf{e} \in L_i(\mathbf{d})$. The multigraded Castelnuovo–Mumford regularity of $\mathcal{F}$ is then the set:*

$$\operatorname{reg}(\mathcal{F}) \coloneqq \left\{ \mathbf{d} \in \mathbb{Z}^r \;\middle|\; \mathcal{F} \text{ is } \mathbf{d}\text{-regular} \right\} \subset \mathbb{Z}^r.$$

The obvious approaches to generalize Theorem 1.1 to a product of projective spaces turn out not to work. For example, the multigraded Betti numbers do not determine multigraded Castelnuovo–Mumford regularity [BHS24, Example 5.1] Despite this we show that part (3) of Theorem 1.1 can be generalized. To do so we introduce the following generalization of linear resolutions.

Definition 1.3. *A complex $F_\bullet$ of $\mathbb{Z}^r$-graded free S-modules is $\mathbf{d}$-quasilinear if and only if F_0 is generated in degree $\mathbf{d}$ and each twist of F_i is contained in $L_{i-1}(\mathbf{d} - \mathbf{1})$.*

Theorem 1.4. [BHS24, Theorem A] *Let M be a (saturated) finitely generated $\mathbb{Z}^r$-graded S-module:*

$$M \text{ is } \mathbf{d}\text{-regular} \iff M_{\geq \mathbf{d}} \text{ has a } \mathbf{d}\text{-quasilinear resolution.}$$

The proof of Theorem 1.4 is based in part on a spectral sequence argument that relates the Betti numbers of $M_{\geq \mathbf{d}}$ to the Fourier–Mukai transform of $\widetilde{M}$ with Beilinson's resolution of the diagonal as the kernel. Recent breakthroughs [HHL24, BE24] understanding resolutions of the diagonal on arbitrary toric varieties mean that there is hope one may be able to generalize the above argument to arbitrary toric varieties. With this in mind, I am interested in pursuing the following question

Building upon our work discussed above – and inspired by previous work on $\mathbb{P}^r$ [BEL91, CHT99, Kod00] – my collaborators and I also studied the asymptotic regularity of powers of ideals on arbitrary toric varieties. In particular, Definition 1.2 can be extended to all toric varieties by letting S be Cox ring of the toric variety X, replacing $\mathbb{Z}^r$ with the Picard group of X, and replacing $\mathbb{N}^r$ with the nef cone of X. My collaborators and I show that the multigraded regularity of powers of ideals is bounded and translates in a predictable way. In particular, the regularity of I^t essentially translates within $\operatorname{Nef} X$ in fixed directions at a linear rate.

Theorem 1.5. [BHS22, Theorem 4.1] *There exists a degree $\mathbf{a} \in \operatorname{Pic} X$, depending only on I, such that for each integer $t > 0$ and each pair of degrees $\mathbf{q}_1, \mathbf{q}_2 \in \operatorname{Pic} X$ satisfying $\mathbf{q}_1 \geq \deg f_i \geq \mathbf{q}_2$ for all generators f_i of I, we have*

$$t\mathbf{q}_1 + \mathbf{a} + \operatorname{reg} S \subseteq \operatorname{reg}\bigl(I^t\bigr) \subseteq t\mathbf{q}_2 + \operatorname{Nef} X.$$

1.2 Top-Weight Cohomology of $\mathcal{A}_g$ The moduli space of (principally polarized) abelian varieties of dimension g, is a smooth variety $\mathcal{A}_g$ (truthfully a smooth Delinge–Mumford stack) whose points are in one to one correspondence with isomorphism classes of principally polarized abelian varieties of dimension g. Concretely, we may view it as the quotient $[\mathbb{H}_g/\mathrm{Sp}(2g, \mathbb{Z})]$ where $\mathbb{H}_g$ is the Siegel upper half-space. Notice this means that $\mathcal{A}_g$ is a rational classifying space for the integral symplectic group $\mathrm{Sp}(2g, \mathbb{Z})$. Similar to the moduli space of curves $\mathcal{A}_g$ has long been studied, but much remains unknown about its geometry. For example, the (singular) cohomology is $\mathcal{A}_g$ is only fully known for $g \leq 4$, with $g = 0, 1$ being relatively easy, $g = 2$ which is a classical result of Igusa [**?**], $g = 3$ due to work of Hain [**?**], and $g = 4$ being deducible from [**?**, **?**]. In fact until recent work by myself and co-authors it was unknown whether $H^{2k+1}(\mathcal{A}_g; \mathbb{Q}) \neq 0$ for some g and k. This was a question posed by Grushevsky. that my recent work answered [**?**].

Building upon the work of Chan, Galatius, and Payne, my co-authors and I developed new methods for understanding a certain canonical quotient of the cohomology of $\mathcal{A}_g$. In particular, our results construct non-trivial cohomology classes in $H^k(\mathcal{A}_g;\mathbb{Q})$ in a number of new cases.

Theorem 1.6. [BBC+24, Theorem A] *The rational cohomology* $H^k(\mathcal{A}_g;\mathbb{Q}) \neq 0$ *for:*

$$(g,k) = (5,15), (5,20), (6,30), (7,28), (7,33), (7,37), \text{ and } (7,42).$$

For broader context, since $\mathcal{A}_g$ is a rational classifying space for $\mathrm{Sp}_{2g}(\mathbb{Z})$ there is natural isomorphism $H^*(\mathcal{A}_g;\mathbb{Q}) \cong H^*(\mathrm{Sp}_{2g}(\mathbb{Z});\mathbb{Q})$. In particular, the above results provide new non-vanishing results for $H^*(\mathrm{Sp}_{2g}(\mathbb{Z});\mathbb{Q})$. However, my work takes advantage of the fact that since $\mathcal{A}_g$ is a smooth and separated Deligne Mumford stack with a coarse moduli space which is an algebraic variety, permitting Deligne's mixed Hodge theory to be applied to study the rational cohomology of these groups. In particular, the rational cohomology of a complex algebraic variety X of dimension d admits a weight filtration with graded pieces $\mathrm{Gr}_j^W H^k(X;\mathbb{Q})$. As $\mathrm{Gr}_j^W H^k(X;\mathbb{Q})$ vanishes whenever $j > 2d$, $\mathrm{Gr}_{2d}^W H^k(X;\mathbb{Q})$ is referred to as the *top-weight* part of $H^k(X;\mathbb{Q})$.) In this way we deduce Theorem 1.6 above as a corollary to computing the top-weight cohomology of $\mathcal{A}_g$ for all $g \leq 7$.

1.3 Broader Impacts from Prior NSF Support

1.3.1 Organizing I organized 11 conferences: *Math Careers Beyond Academia* (50 participants), *M2@UW* (45 participants), *Geometry and Arithmetic of Surfaces* (40 participants), *CAZoom* (70 participants), *Western Algebraic Geometry Symposium* (100 participants), *Gems of Combinatorics I & II* (40 & 30 participants), Spec($\overline{\mathbb{Q}}$) (50 participants), *BATMOBILE (2022)* (20 participants), *Gems of Commutative Algebra I & II* (35 & 40 participants). Additionally, I have organized three special sessions at AMS Sectional Meetings and the Joint Math Meetings. Many of these workshops, namely the *Gems of* series, focused on supporting the next generation of researchers by building research communities and networks.

1.3.2 Math Circles From 2015 through 2018 I was heavily involved in the Madison Math Circle, including 2 years as the lead organizer. As an organizer, I worked to build stronger connections between the Madison Math Circle, local schools and teachers, and other outreach organizations focused on underrepresented groups. This led to weekly attendance to more than double. I also created a new outreach arm of the circle, which visits high schools around the state of Wisconsin to better serve students from underrepresented groups. This dramatically expanded the reach of the Madison Math Circle, and during my final year as an organizer, the circle reached over 300 students. While a post-doc at UC Berkeley I volunteered as a speaker with the Berkeley Math Circle.

1.3.3 Mentoring I have actively sought out ways to mentor undergraduate and graduate students, especially those from generally underrepresented groups. While a graduate student, I led reading courses with three undergraduates. One of these students, an undergraduate woman, worked with me for over a year, during which time I helped her apply for summer research projects. Working with *Girls' Math Night Out* I lead two girls in high school through a project exploring cryptography. During 2018-2019, I mentored 6 first-year graduate students (all women or non-binary students). I also mentored two undergraduate women via the AWM's Mentoring Network.

I advised two summer research projects for undergraduate students. The first of these projects ran virtually during Summer 2021 when 6 undergraduates worked on combinatorial questions related to my work in [BBC+24]. In Summer 2022 I advised an undergraduate student on a research project related to my work on syzygies discussed in Section 1.1. This student is now a graduate student in mathematics and received an NSF GRF. As a postdoc, I began research projects with three graduate students (a majority of whom identify with a generally underrepresented group). These projects have resulted in two pre-prints, with additional projects still ongoing. Throughout the

Spring and Summer of 2022, I did a reading course with a first-year graduate woman on algebraic geometry.

1.3.4 Virtual Mathematics In response to the COVID-19 pandemic and the shift of many mathematical activities to virtual formats, I worked to find ways for these online activities to reach those often at the periphery. During Summer and Fall 2020, I helped with Ravi Vakil's *Algebraic Geometry in the Time of Covid* project. This massive online open-access course in algebraic geometry brought together $\sim 2,000$ participants from around the world. In Spring 2021, I organized an 8-week virtual reading course for undergraduates in algebraic geometry and commutative algebra.

1.3.5 t While a graduate student I co-founded oSTEM@UW as a resource for LGBTQ+ students in STEM, which eventually grew to over fifty active members. As one member said, "It made me very happy to see other friendly LGBTQ+ faces around... Thanks so much for organizing this stuff – it's really helpful". From 2017-2020 I led the campus social organization for LGBTQ+ graduate and post-graduate students, which had over 350 members.

Since Fall 2020 I have organized *Trans Math Day*, an annual one-day virtual conference promoting the work of transgender and non-binary mathematicians. Highlighting the importance of such conferences one student participant said, "I've been really considering leaving mathematics. [Trans Math Day 2020] reminded me why I'm here and why I want to stay. ... If a conference like this had been around for me five years ago, my life would have been a lot better." Trans Math Day regularly has 50 participants.

During 2020-2023 I served as a board member for *Spectra: The Association for LGBTQ+ Mathematicians*, including the inaugural president. I oversaw the growth and formalization of the organization, including the creation and adoption of bylaws, the creation of an invited lecture at the Joint Math Meetings, and a fundraising campaign that has raised over $20,000.

2. The Geometry of Syzygies on Stacks

2.1 N_p-Conditions for Stacky Curves

Research Goal 2.1. *Understand the syzygies of stacky curves.*

Research Problem 2.2. *Let $\mathcal{X}$ be a stacky curve of genus 0 and 1. If $\mathcal{L}$ is a line bundle on $\mathcal{X}$ and $\deg(\mathcal{X}) \gg 0$ what N_p conditions does $R(\mathcal{X} : \mathcal{L})$ satisfy?*

2.2 Green's Conjecture for Canonical Stacky Curves

Research Goal 2.3. *Find and prove an appropriate generalization of Green's conjecture describing the vanishing of the syzygies of the canonical rings of stacky curves.*

Theorem 2.4. [?] **Theorem 8.4.1 Let $\mathcal{X}$ be stacky curve whose coarse space has genus ≥ 1 and let e be the maximum order of the stabilizer groups of the stacky points; then the canonical ring $R(\mathcal{X}; K_{\mathcal{X}})$ is generated by elements in degree at most $3e$ with relations in degree at most $6e$.*

Research Problem 2.5. *Can Theorem 2.4 be refined by considering the existence of certain stacky or weighted linear series on $\mathcal{X}$?*

The proof of Theorem 2.4 builds upon the ideas of Schreyer [**?**] to show the existence of a Gröbner basis for the canonical ring of a stacky curve whose elements have certain degrees. However, Voight and Zureick-Brown do not explicitly construct such Gröbner bases. Instead, they prove the existence of such a Gröbner basis via a delicate induction argument, which allows them to reduce to cases when $\mathcal{X}$ has relatively few stacky points, the stacky points of $\mathcal{X}$ have stabilizers of small order, or the coarse space of $\mathcal{X}$ has small genus. It should be possible to make their arguments explicit for simple stacky curves.

Research Problem 2.6.

With this in mind, one approach toward Problem **??** is to make Voight and Zureick-Brown's arguments explicit for a large number of examples, and then use a computer algebra system like Macaulay2 to compute the minimal graded free resolutions for these examples.

Research Problem 2.7. *Compute the minimal graded free resolution of the canonical ring $R(\mathcal{X}, K_{\mathcal{X}})$ for stacky curves $\mathcal{X}$ where:*

- *the coarse space of $\mathcal{X}$ has genus ≤ 4,*
- *there are at most 4 stacky points, and*
- *the stabilizers of the stacky points have order at most 5.*

Building upon code generously shared by Voight and Zureick-Brown, I have carried out Problem 2.7 for most of the genus zero examples. In many of these computed cases, and as noted in [**?**]*Appendix, the canonical ring is a (weighted) complete intersection. However, there are examples showing that even for small examples the canonical ring of a stacky curve is quite complicated.

Example 2.8. Let $\mathcal{X}$ be a stacky curve with three stacky points whose stabilizers are $\mathbb{Z}/4\mathbb{Z}$, $\mathbb{Z}/4\mathbb{Z}$, and $\mathbb{Z}/5\mathbb{Z}$ whose coarse space has genus zero. The canonical ring $R(\mathcal{X}; K_{\mathcal{X}})$ can be presented as S/I where $S = \mathbb{C}[x_1, x_2, x_3, x_4, x_5]$ where $\deg(x_1) = 3$, $\deg(x_2) = \deg(x_3) = 4$, and $\deg(x_4) = \deg(x_5) = 5$ and I is an ideal generated by 7 homogenous polynomials of degrees $8, 8, 9, 9, 10, 13, 13$. The minimal graded free resolution of $R(\mathcal{X}; K_{\mathcal{X}})$ is shown below (with the degrees suppressed for brevity):

$$0 \longleftarrow R(\mathcal{X}; K_{\mathcal{X}}) \longleftarrow S \longleftarrow S^{\oplus 7} \longleftarrow S^{\oplus 20} \longleftarrow S^{\oplus 30} \longleftarrow S^{\oplus 22} \longleftarrow S^{\oplus 6} \longleftarrow 0.$$

The approach of using large-scale computations to find conjectures concerning syzygies is very much in the same spirit as my prior work on the syzygies of surfaces [**?**, BCE+22]. While the precise techniques need to address Problem 2.7 will likely be different much of the general framework of organizing, distributing, and publicizing such computations will likely be similar.

2.3 Asymptotic Syzygies of Weighted Projective Stacks

2.4 Computations with Toric Stacks

Research Problem 2.9. *Develop a robust software package for Macaualy2 allowing for computations with toric stacks.*

3. Geometric & Algebraic Measures of Complexity on Toric Varieties

3.1 Asymptotic Regularity of Powers of Ideals and Ideal Sheaves

Research Goal 3.1. *Let X be a smooth projective toric variety with S it's $\operatorname{Pic}(X)$-graded Cox ring. Understand the asymptotic behavior of $\operatorname{reg}(I^p)$ and $\operatorname{reg}(\mathcal{I}^p)$ where $I \subset S$ is an $\mathbb{Z}^r$-graded ideal and $\mathcal{I} \subset \mathcal{O}_X$ is an ideal sheaf.*

Conjecture 3.2. *Let X be a smooth projective toric variety and $\mathcal{I} \subset \mathcal{O}_X$ an ideal sheaf. With the notation as above:*

$$\lim_{p \to \infty} \frac{\operatorname{reg}(\mathcal{I}^p)_{\mathbb{R}}}{p} = \left\{ L \in\in \operatorname{Pic}(X)_{\mathbb{R}} \;\middle|\; \pi^*(L) - E \textit{ is nef on } \tilde{X} \right\}.$$

Research Problem 3.3. *Prove Conjecture 3.2.*

Research Problem 3.4. *Let X be a smooth projective toric variety with S it's $\operatorname{Pic}(X)$-graded Cox ring. If $I \subset S$ is a $\mathbb{Z}^r$-graded ideal bound the number of minimal generators of $\operatorname{reg}(I^p)$ as a function of p in terms of invariants of I?*

3.2 A Generalized Eisenbud–Goto Theorem for Toric Varieties

Research Problem 3.5. *Let X be a smooth projective toric variety with S it's* $\mathrm{Pic}(X)$*-graded Cox ring and B its irelevant ideal. Let M be a* $\mathrm{Pic}(X)$*-graded S-module. Find the correction generalization of d-quasi-linear resolution such that a* $\mathrm{Pic}(X)$*-graded S-module M is d-regular if and only if the truncation $M_{\geq d}$ has a d-quasilinear resolution.*

Research Problem 3.6. *Let X be a smooth projective toric variety with S it's* $\mathrm{Pic}(X)$*-graded Cox ring and B its irelevant ideal. Let M be a* $\mathrm{Pic}(X)$*-graded S-module. Find a characterization for when the truncation $M_{\geq d}$ for $d \in \mathrm{Pic}(X)$ has a linear resolution in terms of vanishing of the local cohomology groups $H^i_B(M)_t$.*

3.3 Regularity on the D_{Cox} Category

3.4 Bondal-Thompson Resolutions of Kernel Bundles

3.5 The Free Resolution of $\overline{\mathcal{M}}_{0,n}$

Research Problem 3.7. *Compute the multigraded Castelunovo–Mumford Regularity of $\overline{\mathcal{M}}_{0,n} \subset$??*

Research Problem 3.8. *Compute the minimal graded free resultion of the defining ideal I of $\overline{\mathcal{M}}_{0,n} \subset$??*

4. Broader Impacts

4.1 Organizing As discussed previously, I have organized 10+ national/international conferences, and I plan to continue organizing conferences at roughly this pace. Currently, I am organizing two research conferences in algebraic geometry *Algebraic Geometry Northeastern Series (AGNES)* and *BATMOBILE* both occurring in late 2024. I am also in the early stages of organizing an international workshop with Diane Maclagan on multigraded homological algebra and toric geometry, which I hope will occur in 2025. Tyler Kelly, Mike Hill, and I are also in the planning stages of organizing a satellite conference for LGBTQ+ mathematicians for the 2026 ICM in Philadelphia.

The"GEMS" workshop series that I co-founded sought to build a diverse community of mathematicians to address gender equity in the mathematical community from new perspectives. In particular, these conferences are indented to supplement the affinity group model of the *Women in...* conference series by promoting broad conversations and networks. Going forward I am interested in expanding these "GEMS" workshops to other areas of mathematics and creating cross-field discussions that broaden the standard notion of gender equity in mathematics. In particular, there is a planned follow-up to *GEMS in Commutative Algebra* scheduled for Fall 2025, and I am tentatively planning a version of *GEMS in Algebraic Geometry* for 2026. As more of these conferences occur, I have plans to try to work with mathematicians in other fields to organize their own version of *GEMS in...* conferences in their fields.

When organizing these conferences I paid particular attention to making them as inclusive of women and non-binary researchers as possible. For example, I designed the registration form to be thoughtful of the concerns of transgender researchers and highlighted the locations of single occupancy and ADA-compliant restrooms. The importance of such efforts was highlighted by the following comment I received from a participant, "I just wanted to thank you for making this workshop inclusive for people with all gender identifications. ... I have always felt out of place when I participated in conferences/workshops for women when they do not specify that non-binary people are welcome ... I really appreciate those questions you put in the registration form. It means a lot to me." I would like to work with various professional societies (AMS, AWM, NAM, Spectra, etc.) and research institutes (SLMath, ICERM, IAS, etc.) to produce a guide of best practices for organizing welcoming and inclusive mathematics conferences. By doing this I hope to make it easier for others to organize welcoming events.

4.2 National & International Advocacy. I have worked with the Executive Committee of the *Association for Women in Mathematics* to consider ways they could expand their support of women and non-binary mathematicians. I hope to continue this work and find further ways the AWM can support both women and non-binary mathematicians. Since Winter 2023 I joined SLMath's (formerly MSRI's) *Committee on Women in Mathematics.* I plan to remain on this committee until 2028 when my term is set to expire. In this role, I am hoping to organize a yearly retreat at SLMath that brings together small groups of 5-8 women (and underrepresented individuals more generally) who have been implementing innovative ideas promoting equity in mathematics and brainstorm ways to scale their efforts and effect radical positive change for women in mathematics.

4.3 A More Inclusive Learning Community. Going forward I am excited to continue my work supporting LGBTQ+ students, and would love to continue building organizations to do so. In particular, given the amazing successes of programs like MSRI-UP and the EDGE Program, I would love to organize a summer REU program specifically aimed at supporting and promoting LGBTQ+ mathematicians. Further, I am in the early stages of planning a mentorship program to help guide LGBTQ+ undergraduates through the process of applying to graduate programs in mathematics and helping young LGBTQ+ graduate students establish themselves. The plan would be to break participants into groups with each group having LGBTQ+ mathematicians at various career stages, thus allowing participants to exchange advice, find support, and build mentoring networks.

4.4 Mentoring. As a new faculty member at Dartmouth College I hope to continue and expand the mentoring I do with students, continuing a focus on mentoring students from underrepresented groups. Since arriving on campus in the late Summer I have begun mentoring one graduate student and one undergraduate woman. While still early it seems likely that I will end up co-advising the graduate student and advising the undergraduate's senior thesis. I am also working with the college's AWM Student Chapter and the Out in STEM Student Chapter.

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